

INT332 – DevOps

Unit 3 – Week 7

Monolithic vs Microservices Architecture

Subject Information

Subject: INT332 DevOps

Unit: 3

Week: 7

Topic: Monolithic vs Microservices Architecture

Learning Outcomes

- Understand monolithic architecture.
- Understand microservices architecture.
- Compare both architectures.
- Learn scalability, isolation, agility, and API Gateway concepts.

1. Monolithic Architecture

Monolithic architecture is a traditional software design approach in which all components of an application are combined into a single codebase and deployed as one unit. User interface, business logic, and database access layers are tightly integrated. Examples include many legacy enterprise applications where every feature is packaged together in a single deployable file.

Characteristics of Monolithic Architecture

- Single codebase
- Single deployment unit
- Shared database
- Tightly coupled components
- Easier initial development

Advantages of Monolithic Architecture

1. Simpler development for small applications.
2. Easier testing during early stages.
3. Straightforward deployment process.
4. Less operational complexity.

Disadvantages of Monolithic Architecture

1. Difficult to scale individual features.
2. Large codebases become harder to maintain.
3. A single failure can impact the entire application.
4. Slower development as teams grow.
5. Technology stack changes become difficult.

2. Microservices Architecture

Microservices architecture divides an application into multiple small services. Each service performs a specific business function and communicates with other services through APIs. Services can be developed, deployed, and scaled independently.

Characteristics of Microservices Architecture

- Independent services
- Independent deployment
- Loose coupling
- Service-specific databases
- Technology flexibility
- API-based communication

Advantages of Microservices Architecture

1. Better scalability.
2. Faster deployments.
3. Improved fault isolation.
4. Independent development teams.
5. Easier adoption of new technologies.

Disadvantages of Microservices Architecture

1. Higher operational complexity.
2. More infrastructure requirements.
3. Monitoring becomes challenging.
4. Increased network communication.
5. Data consistency challenges.

Monolithic vs Microservices Comparison

Deployment: Monolithic = Single Unit, Microservices = Independent Services.

Scalability: Monolithic = Entire application, Microservices = Individual services.

Fault Isolation: Monolithic = Low, Microservices = High.

Maintenance: Monolithic = Difficult as application grows, Microservices = Easier due to smaller services.

Team Collaboration: Monolithic = Limited, Microservices = Better team autonomy.

Scalability

Scalability refers to the ability of a system to handle increasing workloads. In monolithic systems, the entire application must be scaled. In microservices, only heavily used services need additional resources, reducing costs.

Isolation

Isolation means that failure in one component does not affect others. In microservices, if the payment service fails, the product catalog may continue working. This improves system reliability.

Agility

Agility refers to the speed at which teams can develop, test, and deploy changes. Microservices enable multiple teams to work independently, increasing development speed and innovation.

API Gateway

An API Gateway acts as a single entry point for client requests. Responsibilities:

- Request routing

- Authentication and authorization

- Load balancing

- Rate limiting

- Monitoring and logging

API Gateways simplify communication between clients and multiple microservices.

Exam-Oriented Points

- Monolithic applications are deployed as a single unit.
- Microservices are deployed independently.
- Microservices provide better scalability and fault isolation.
- API Gateway is a central entry point in microservices architecture.
- Modern cloud-native applications commonly use microservices.

Viva Questions

1. What is monolithic architecture?
2. What is microservices architecture?
3. Differentiate between monolithic and microservices systems.
4. What is scalability?
5. What is an API Gateway?
6. Why are microservices considered more agile?

Summary

Monolithic architecture is simple and suitable for smaller applications, while microservices architecture offers better scalability, fault isolation, and agility for large systems. Organizations increasingly adopt microservices to support rapid development and cloud-native deployments.